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To cite this article: Jason Freeman, Shaoduo Xie, Takahiko Tsuchiya, Weibin Shen, Yan-Ling Chen & Nathan Weitzner (2015) Using massMobile, a flexible, scalable, rapid prototyping audience participation framework, in large-scale live musical performances, Digital Creativity, 26:3-4, 228-244, DOI: [10.1080/14626268.2015.1057345](https://doi.org/10.1080/14626268.2015.1057345)

To link to this article: <https://doi.org/10.1080/14626268.2015.1057345>



Published online: 28 Jul 2015.



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Using massMobile, a flexible, scalable, rapid prototyping audience participation framework, in large-scale live musical performances

Jason Freeman, Shaoduo Xie,
Takahiko Tsuchiya, Weibin Shen,
Yan-Ling Chen and Nathan Weitzner

Center for Music Technology, Georgia Institute of Technology

jason.freeman@gatech.edu; shaoduo.xie@gmail.com; takahiko@gatech.edu;
weibenizer@gmail.com; yanlingchen0815@gmail.com; nweitzner@mat.ucsb.edu

Abstract

massMobile is a client–server system for mass audience participation in live musical performances using existing wireless network infrastructure, audience members' own mobile computing devices (such as smartphones) and a cloud-based server to enable rapid prototyping, iterative design, and plug and play deployment of participatory systems under constraints common to live events and venues. In this paper, we specifically focus on technical and design issues related to large-scale participatory performances, exploring (a) what technical designs provide robust support; (b) what interfaces and mappings can be effective and (c) what strategies can effectively focus and guide participation. We first briefly address previous work in audience participation in live performance, then describe massMobile's system design and implementation with a focus on new functionality to support robust use at scale. Finally, we discuss two sets of events using massMobile in large-scale performance contexts: with an athletic band at sporting events and with a jazz combo in traditional concert venues.

Keywords: music, audience participation, cloud computing, drawing, collaboration

1 Introduction

We have long been interested in participatory interfaces in live musical performances to create a sense of shared community among audience members and a dynamic of risk, spontaneity, excitement and uniqueness at live concerts (Freeman and Godfrey 2010; Weitzner et al. 2012). In doing so, we push against current tendencies for live musical events to seek to reproduce the perfection of studio recordings through a reliance on prerecorded content and intensely pre-choreographed musical expressivity. Instead, we challenge audiences and onstage performers to collaborate to create a musical outcome truly unique to each performance, blurring the lines between composition, performance and listening (Weitzner et al. 2013).

As smartphones become increasingly ubiquitous, powerful and connected, they present a unique opportunity to facilitate interactive and expressive collaborative experiences in live musical performances. Compared to systems incorporating dedicated hardware, such as compu-

ter vision, smartphone-based systems can reduce development time, enabling a rapid iterative design loop, and they can reduce deployment time, easing adoption under constraints common to live performance venues. massMobile, the client–server audience participation framework we have been developing since 2011 (Weitzner et al. 2013), leverages these opportunities, utilizing existing wireless network infrastructure, audience members' own mobile computing devices and a cloud-based server to enable rapid prototyping, iterative design and plug and play deployment of participatory systems. The framework both supports our own artistic work and is available as a free tool for other artists to incorporate into their own explorations.

The ubiquity of smartphones also makes it possible to scale such participatory performances to large numbers of audience participants. In this paper, we specifically focus on technical and design issues related to large-scale participatory performances that may include hundreds of simultaneously participating audience members, exploring (a) what technical designs provide robust support; (b) what interfaces and mappings can be effective and (c) what strategies can effectively focus and guide participation. We first briefly address previous work in audience participation in live performance, then describe massMobile's system design and implementation with a focus on new functionality to support robust use at scale. Finally, we discuss two sets of events using massMobile in large-scale performance contexts: with an athletic band at sporting events and with a jazz combo in traditional concert venues.

2 Background

Taxonomies and histories of interactive music systems are well represented in prior literature (e.g. Rowe 1992; Jorda 2007; Drummond 2009). We can consider audience participation systems through two distinct lenses. The field of network music (e.g. Barbosa 2003; Weinberg 2005) considers interactive musical systems from the perspective of musical participants and the flow of data among them, providing a means to describe and

analyze the roles and connections among audience participants, onstage performers and technology. The esthetic practice of open-form composition (e.g. Brown 1999; Thorington 2005; Harries 2013), in contrast, offers a foundation for considering the role of a (pre-composed) musical structure, improvisation and chance in the performance of music employing such a networked topology.

Our interest in audience participation systems with massMobile focuses specifically on live musical performances, as opposed to interactive sound installations (e.g. Schacher 2008) or shared sonic environments (e.g. Barbosa 2003; Bryan-Kinns and Healey 2004). In a prototypical participatory performance, a large audience is co-located in time and place, creating sound and/or controlling the sound that others produce in concert. The audience may be seated, as in a traditional concert venue (e.g. Oh and Wang 2011), or standing or even dancing, as in a club (e.g. Feldmeier and Paradiso 2007). One or more musicians on stage may also play a role in creating sound and/or facilitating the performance (Lee and Freeman 2013; Wyse 2015).

Network design is usually constrained by the large number of simultaneous inputs from audience participants and the primacy of a single musical result that all participants hear. Designers of such systems must also contend with the norms of the performance context and expectations of participants, who may be used to more subtle forms of participation such as applause or dancing and may embrace, reject or ignore opportunities to engage more actively in music production (Dannenberg and Bates 1995).

Participatory performances need not depend on technology. For example, in *Moths* (Hasse 1986), an onstage conductor leads a seated audience of whistlers guided by a graphical score to perform the piece. The introduction of technology, though, makes possible new approaches to connecting audiences, performers and musical processes at scale and of recasting the roles of each participant and the ways in which they communicate and collaborate.

Computer vision has long been employed in interactive music systems, particularly with

dance and movement (e.g. Rokeby 1998; Wechsler, Weiss, and Dowling 2004). In large-audience gaming applications, computer vision and motion tracking have used similar techniques to follow the small-scale movements of audience members within their seats (Maynes-Aminzade, Pausch, and Seitz 2002), the location of objects shared among audience members (Bregler et al. 2005) or binary responses (Carpenter and Carpenter 1999). Similar techniques have been employed in live musical performances. For example, in *Flock* (Freeman and Godfrey 2010), computer vision tracks the locations of participants within the performance space. Computer vision systems usually require ample time to setup and calibrate in performance spaces—a requirement that cannot often be met in real performance venues. Most such systems are also limited in the resolution and dimensionality of data they detect from large audiences.

In recent years, musicians and researchers have increasingly turned to smartphones in participatory systems. Golan Levin, in *DialTones*, used audience mobile phones as a distributed array of loudspeakers (Traub 2005), an approach that has been echoed in more recent works (e.g. Wyse 2015). Other recent works (e.g. Lee and Freeman 2013; Schnell et al. 2015) similarly diffuse sound across audience mobile device speakers, but also give each audience member the agency to trigger, repeat or create musical material within constraints. Others have moved away from the sound diffusion model, using text messaging (Dahl, Herrera, and Wilkerson 2011; Kim 2011) or control interfaces (Roberts and Hollerer 2011; Allison, Oh, and Taylor 2013; Hindle 2013) to give audience members collective influence over centrally generated electroacoustic sound.

In massMobile, we take advantage of ubiquitous mobile computing devices for real-time, large-scale audience participation with multidirectional communication. Following the example set by more generalized interactive music frameworks such as EyesWeb (Camurri et al. 2007), we sought to create a flexible standards-based framework built to enable artists and researchers to rapidly

and iteratively design and deploy robust systems. Inspired by prior work and driven in particular by the deployment challenges and usability issues that pervade large-scale participatory performances, our overarching design goals for massMobile were to (a) allow rapid development of participatory systems by the user/artist, (b) enable a seamless plug and play deployment of those systems in live concert venues and (c) enable instant access to participation for audience members using their own mobile computing devices without any special network configuration or application install. In designing the system and balancing issues such as latency and interface design, we always favored decisions that served these goals. These goals also supported our iterative approach to exploring our core research questions about effective technical designs, interfaces and mappings, and strategies to focus and guide participants in large-scale participatory systems for live musical performance.

3 Framework design and implementation

massMobile is designed as a client–server software framework (Figure 1). It consists of (a) a mobile-optimized web client for use by audience members on any computing device (iOS, Android, laptop, etc.) running a recent browser and connected to the Internet via Wi-Fi, 3G or 4G; (b) a Java application programming interface (API) that enables musicians to rapidly develop software to poll raw and aggregate audience data from a cloud-based server to obtain and analyze audience input, respond accordingly and dynamically reconfigure the audience web client interface and (c) an offsite cloud-based service that processes requests from the web client and the Java API to collect audience input, report raw and aggregate audience data, and update user interface configuration.

As much as possible, we have sought to use industry-standard and open-source tools (such as Node.js, MongoDB and Google Web Toolkit (GWT)), languages (such as Java) and protocols

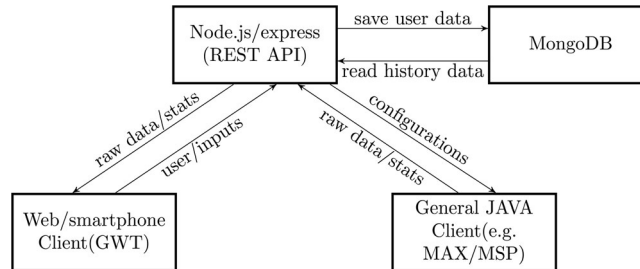


Figure 1. Architecture of the massMobile framework.

(such as REST) to help make massMobile robust, portable and extensible.

3.1 Server, database and deployment

3.1.1 Server and database

Our original implementation of massMobile ran on a single shared server with a Java/MySQL backend (Weitzner et al. 2013). This presented two challenges to us: the server's limited capacity made it difficult to scale beyond performances with 50–100 simultaneous participants, and its highly structured database design made it time-consuming to add server-side code and database structures to support new user interfaces. This made it hard to deploy massMobile in the large-audience performance situations we sought and hampered our ability to rapidly design and test new interaction paradigms.

Our current server is instead implemented using Node.js and Express, with the MongoDB NoSQL database. This facilitates data sharing and analysis in near real time at scale and provides a mechanism to archive audience input for later analysis. This light-weight, flexible architecture has enabled more rapid development and vastly improved response times and scalability in performance.

3.1.2 Deployment

Instead of hosting massMobile on a single shared server, we now host massMobile using PaaS (platform as a service) to enable greater scalability by instantly ramping up capacity to meet demand. We are also able to distribute massMobile to other artists without the burden of hosting their projects ourselves: they can easily set up their

own PaaS hosting for massMobile according to our deployment guide. We currently use Heroku, a popular PaaS that supports Node.js and MongoDB and has a free usage tier that has easily accommodated the scaling demands of even the larger projects we describe below.

3.2 Client

3.2.1 Web and smartphone client

In early implementations of massMobile, we developed dedicated native applications for iOS and Android. The current massMobile smartphone client is instead realized as a web application in order to simplify development and deployment across multiple platforms without a discernible effect on application performance. It is implemented with GWT, a popular framework in which developers author Java code that is then cross-compiled to highly optimized JavaScript.

During a performance, audience members access the client through a short URL or Quick Response code that points to a full URL that includes configuration parameters, such as the desired project in which to participate, as query string parameters.

Currently available interface widgets for the client include text entry, voting, sliders, drawing and a “2-D” slider for simultaneous x and y inputs. All widgets not only transmit audience data to the server but also have options to visualize individual and/or aggregate responses of other audience members within the interface.

3.2.2 Java API

massMobile is designed to be used in a variety of performance contexts, so we provide a generalized

API in Java to easily connect and communicate with the server at a high level of abstraction, rather than a single, monolithic application to drive performances. massMobile includes sample applications demonstrating the API's use in pure Java applications.

3.2.3 Max/MSP client

We created a Java object and sample patches for Max/MSP to provide access to the full Java API from within Max. We also developed an authoring tool in Max for creating, configuring, editing and testing project configurations, as well as examples of raw and aggregate data polling and dynamic project configuration. We created these abstractions as a convenience to Max/MSP users. Max/MSP is not required to use massMobile. The Java API can be used directly or via another Java-friendly environment such as processing. Figure 2 shows a screenshot of the Max configuration patch.

3.3 Data analysis tool

Our iterative approach to designing interfaces and mappings for massMobile projects, along with our research questions about effective interfaces and mappings and effective strategies to focus and guide participation at scale, necessitated frequent review of test events and public performances to inform future development. While audience feedback and the musical and visual content of the event itself provided valuable formative data, we also sought insights from the participation data itself to track participation levels (e.g. how many people participated and for how long) and interaction trends (e.g. variations in input density and the convergence or divergence of input data).

Since all audience interaction with massMobile is anonymously stored in the MongoDB database, we developed an analysis script, written in Python, to analyze these data and report aggregate

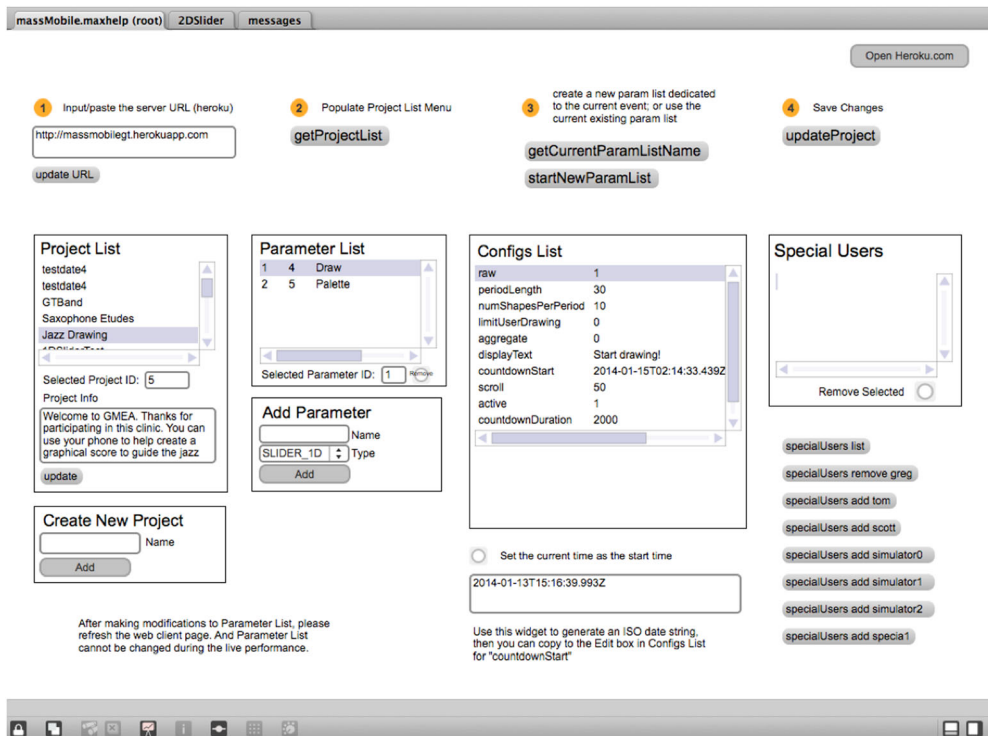


Figure 2. The Max/MSP configuration patch for massMobile projects.

information. The script extracts data such as the number of users, interface input events, and aggregate average and median values, as well as individual raw input data for particular (anonymous) users or time ranges. Results are exported to a comma-separated values file to permit easy analysis and visualization within any spreadsheet or statistics program.

4 massMobile projects at scale

The technical design described in Section 3 provided a robust foundation for designing and deploying large-scale participatory performances. In 2013, we used massMobile in two different large-scale performance contexts: TeamWork, a collaboration with the Georgia Tech Pep Band; and Sketching, a collaboration with the Georgia Tech Jazz Ensemble. With each project, we iteratively improved the interaction design over multiple private test events and public performances, guided by informal feedback from audience members and performers, quantitative participation data from the massMobile database and analysis of each resulting musical performance.

4.1 TeamWork

In TeamWork (Figure 3), an athletic pep band plays their team's fight song at a sports game, while the fans in the stadium use their mobile phones to continuously change the tempo and the dynamics of the performance. This creates a novel way for fans to engage with the band and with each other as they hear their team's familiar music. In recent decades, large sports events have increasingly turned toward playing prerecorded music during game timeouts, even when a live band is present in the stadium. We wanted to create a musical interaction between fans and musicians that put forward an alternate vision for the role of live music at athletic events and the ways in which music can engage fans.

We created TeamWork for the Georgia Tech Pep Band. It was designed for the band to use during halftime at a series of basketball games at the McCamish Pavilion, the university's new basketball arena. We used massMobile to create a web

client for audience participation and a Max/MSP patch to generate a live scoreboard display based on audience input.

From the perspective of network music topologies (Weinberg 2005), TeamWork is a symmetric, one-way synchronous flower topology. From the perspective of open-form works (Brown 1999), TeamWork is heavily dependent on a pre-composed score familiar to both audiences and performers, with real-time transformations to that score that extend beyond traditional limitations on artistic interpretation without losing the identity of the original music. The structural simplicity of these design approaches was a deliberate effort to limit the complexity of an experience that was large scale in size but short scale in duration.

4.1.1 Interface and mapping

Our initial design work focused on identifying effective interfaces and mappings for this context, beginning with two exploratory rehearsals with the ninety musicians that comprise Tech's rotating pep bands. We tried out several interface designs with the band, including a single slider, checkbox voting and the 2-D slider interface discussed below, and we also tried out different strategies for mapping this input to musical qualities such as orchestration and tempo. With each experiment, half the band performed and the other half participated as if they were fans. At the end of each rehearsal, we led an open discussion with the band to gather feedback on each approach. The group reached consensus that the 2-D slider interface, with dimensions mapping onto tempo and dynamics, was the most effective interface and mapping. It was simple enough to understand without explanation, and the mappings led to substantial changes in the music that were easy to hear and easy to play.

The 2-D slider interface (Figure 4) is a plane coordinate system where each point represents two values and the x and y axes are clearly labeled. Audiences voted continuously for their desired tempo and loudness to be played by the band, updating their vote as much as they



Figure 3. Performance of TeamWork at Georgia Tech's McCamish Pavilion. Photograph by Yan-Ling Chen. Reproduced with permission.

wished over the course of the event. The Max/MSP patch generated a display for the arena's scoreboard (Figure 5) that showed the same axes as the client along with average and standard deviation values displayed as an ellipse. The pep band's drummer adjusted his tempo and loudness to the data on the scoreboard, and other players followed his tempo while looking at the scoreboard or the conductor for dynamics.

4.1.2 Guiding participation at scale

TeamWork was performed during halftime in ~2-minute events at 4 men's and women's basketball games in January and February 2013, with official stadium attendance ranging from 624 to 7,128. (The arena's capacity is 8,600). We used these four events to focus on effective mechanisms for focusing and guiding audience participation at scale. At the conclusion of each performance, we

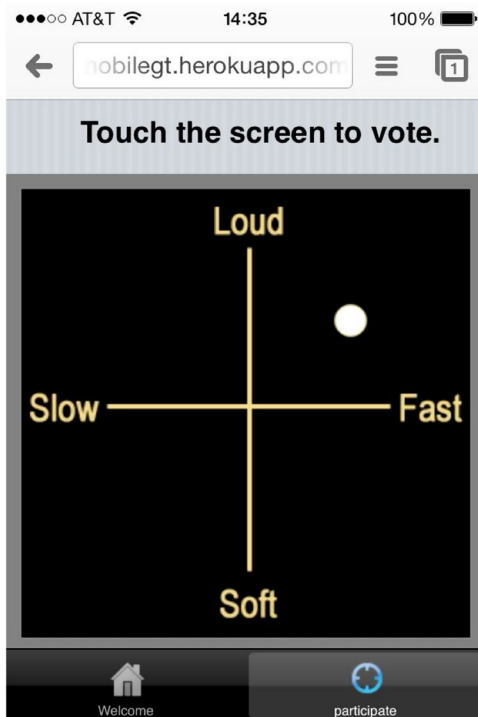


Figure 4. The mobile web client for *TeamWork* for fan participation.

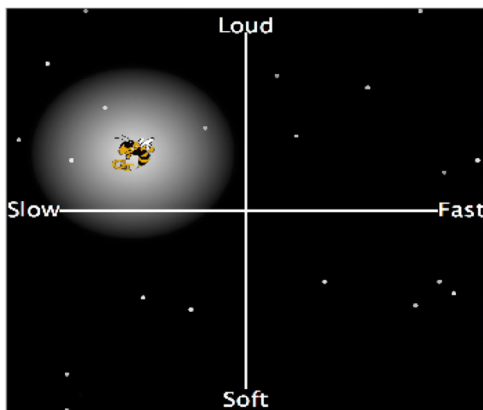


Figure 5. The scoreboard display for *TeamWork*.

analyzed audience input data, the musical outcome itself and audience feedback (collected through free-response text in a mobile phone-based survey and through informal conversations) to drive the next design iteration.

Reconsidering *TeamWork* in the context of a synchronous flower topology (Weinberg 2005), digital network communication is primarily unidirectional (mobile client to cloud server) but the interaction itself is bidirectional, with visual feedback to participants (here via the stadium scoreboard) critical to supporting interaction (Jordà 2003). We expected that simply displaying average and standard deviation data on the scoreboard would not be enough to afford individual agency and collective behavior, but we also believed that displaying the raw data points of hundreds or thousands of users was too computationally expensive and too visually dense.

At our 3rd February performance—the first with a large crowd—we used a compromise solution in which the scoreboard displayed the average and standard deviation values along with raw data from a handful of pre-selected “special” participants. Our hope was that the remaining participants would flock around the points of the “special” participants, thus mimicking the self-organizing dynamics of a small-scale interaction with a large crowd. But the “special” points exerted little influence, so fan input followed a fairly uniform distribution, with the average point hovering around the center of the scoreboard display and the band never venturing away from moderate tempo and dynamics, as shown in Figure 6(b). Fans complained that “it averaged out, so nothing happened” and suggested that they “need better feedback as to how my participation affected the performance, if at all”.

The 3rd February performance, though, also suggested that our compromise on scoreboard display was not actually necessary, since fewer fans participated than we had expected (Table 1). For the remaining two performances, we decided to display all individual fan input from the immediately preceding five seconds instead of input from a handful of pre-selected participants. These dots faded gradually over the course of five seconds so that the newest votes were always most visually prominent. We also invited the team’s mascot to take on the role of influencing audience actions in a way that the “special” participants had been unable to do. We prepared four

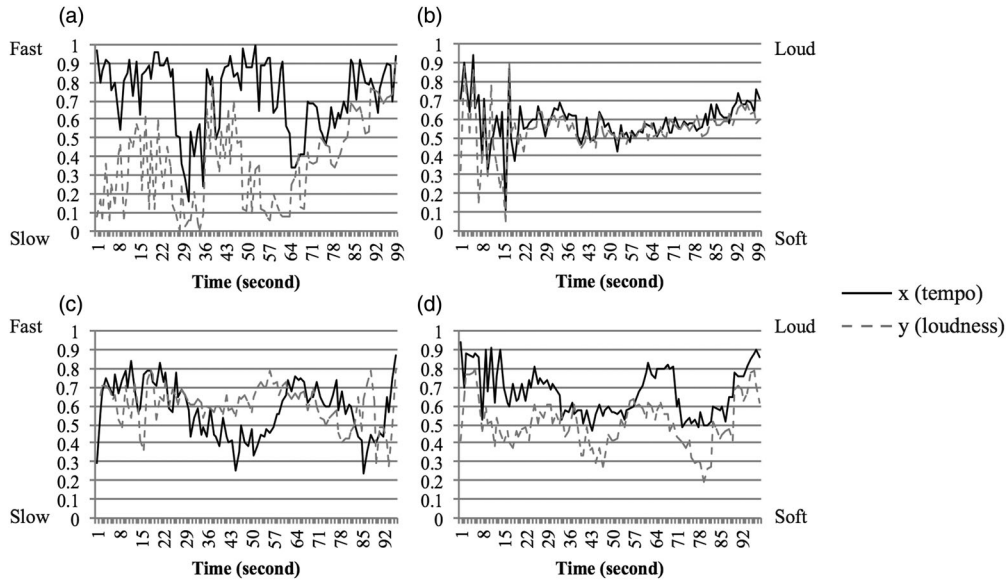


Figure 6. A comparison of average input values over time for each of the four performances of TeamWork: (a) 31 January, (b) 3 February, (c) 27 February and (d) 28th February.

	31 January Women's	3 February Men's	27 February Men's	28 February Women's
No. of users (fans)	11	108	61	19
No. of inputs (votes)	874	5,202	2,883	3,272
Avg. no. of inputs per fan (votes)	75	48	47	172
Med. no. of inputs per fan (votes)	8	25	23	78
Attendance (fans)	624	7,128	6,962	823
Participation rate (%)	1.9	1.8	1.1	3

Table 1. Fan participation statistics from four live performances of TeamWork.

Note: The participation rates are approximate values based on the official attendance reported by the McCamish Pavilion. Actual attendance is typically lower than official reports, so actual participation rates were likely somewhat higher.

signs with “loud”, “soft”, “fast” and “slow” written on each, and asked the mascot to run around the court to try to guide participants’ voting in particular directions. Finally, we stored some history values of the averages, scaled the average point among those past values and showed the scaled average point instead of the true average. This method exaggerated even small deviations from the exact center of the screen while keeping the interpretation meaningful, emphasizing the initial stages of a trend in a particular direction and therefore encouraging other fans to follow that trend.

These changes had a significant impact on the remaining performances. As Figure 6 shows, the average values varied much more significantly and meaningfully in the final two performances (Figure 6(c) and 6(d)) than in the second (Figure 6(b)) and the fans were collectively able to exert substantial influence on the music.

4.1.3 Discussion

TeamWork demonstrated that our technical design for massMobile worked robustly at scale with zero in-venue setup. It also suggested that at scale, X/Y interfaces provide intuitive mechanisms for bidir-

ectional communication between mobile clients and a central server when each axis is mapped to substantial and clearly audible musical change and when each audience input point is clearly displayed back to all participants. Finally, the project showed that designs in which participation is influenced by a charismatic leader (in this case, the team mascot) enable the audience to more effectively effect musical change.

We are disheartened, however, by the extremely low participation rates. Perhaps these rates would improve if TeamWork became a regular part of the game experience and if it were featured at a more prominent point during the game. But there seems to be an upper limit on interest in participation. We spoke informally with several fans after the games. While fans were universally excited about TeamWork, most had not participated because they were talking, eating a snack or otherwise focused on off-court activity, viewing timeouts and halftime as “down time” to be used for other purposes. Given this barrier, we have increasingly turned our attention back toward more traditional performance contexts in

which audiences come to engage with music as a primary, not secondary, activity. That was exactly our approach with Sketching, which we now discuss in detail.

4.2 Sketching

In Sketching (Figure 7), which we presented in two performances with the Georgia Tech jazz combo in spring 2013, the ensemble performs an existing jazz standard (in this case, *So What* by Miles Davis). Upon finishing the head of the song, performers improvise based on a graphical score that is collaboratively drawn by the audience during the performance.

From the perspective of network topologies, Sketching is similar to TeamWork, employing a synchronous, bidirectional flower topology (Weinberg 2005). And while data are similarly organized in an X/Y format, the data exchanged over this network are more complex in content and structure than in TeamWork. Individual data points are associated with a customizable shape and have greater persistence across time, and

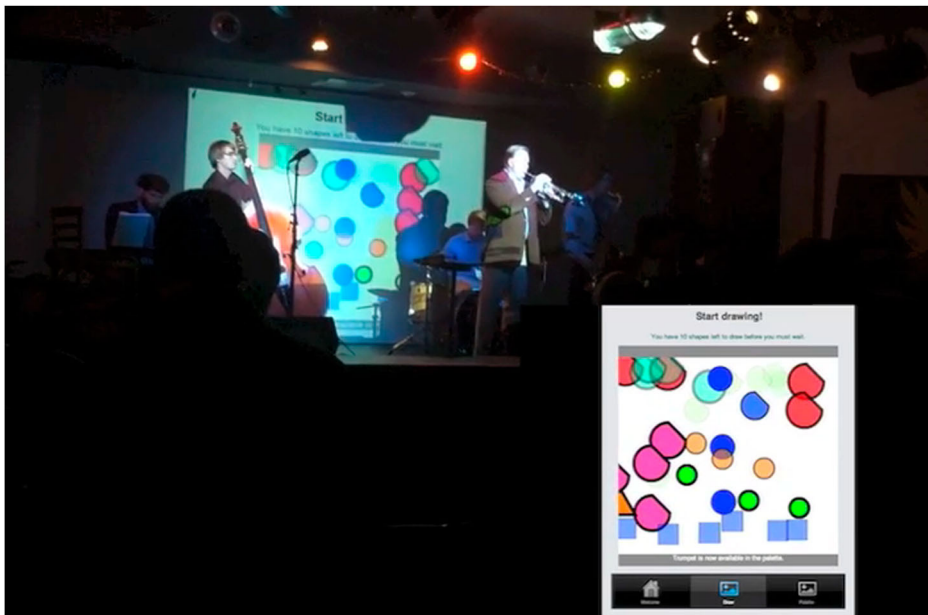


Figure 7. Performance of sketching by the Georgia Tech jazz combo at Under the Couch. Photograph by Yan-Ling Chen. Reproduced with permission.

users can create multiple simultaneous shapes that represent future events.

From the perspective of open form, *Sketching* borrows the harmonic structure and introductory melody from an existing piece of music but leaves most everything else to be determined in real time, such that not only the music's interpretation but the score itself is created in performance. This approach builds on a long tradition of open and graphical scores in experimental musical traditions bridging jazz and the avant-garde (Stockhausen 1957; Zorn 1995; Brown 1999). In most of these examples, though, the unusual scores remain invisible to the audience, who rarely understand the nature of the score, its role in the performance, or the dividing line between composition and expressive interpretation. *Sketching* therefore simultaneously draws from a tradition of projects using technology to connect drawing to musical creation (Lohner 1986; Paul and Werner 2003; McAllister, Alcorn, and Strain 2004; Pitaru 2005) both to reveal the novel score and collaboration to the audience and to challenge them to become involved in the dynamic, collaborative construction of this score.

From both the perspective of network music and the open form, *Sketching* is more structurally complex in its design than *TeamWork*. Because *Sketching* was a longer duration performance taking place in a context focused on music, we were able to introduce the system to the audience, to practice with them and to give them ample time to explore it during the concert. *Sketching* also had explicit pedagogical goals: we sought to help audiences understand more about how musicians improvised through their role in creating a score to guide improvisation; and we hoped that score would provide a useful structural foundation to musicians with limited experience in improvisation.

4.2.1 Interface and mapping

Our iterative approach to creating the interface for *Sketching* involved a series of experiments initially conducted internal to our team and gradually broadened to include additional musicians and

participants. We initially focused on freeform drawing, but we quickly discovered that audience input tended toward comical gestures or words rather than meaningful collaboration, and that the widely varied drawings were difficult for musicians to interpret. We made several attempts, without success, to graft structure onto freeform drawing through mechanisms such as staff lines, scrolling and segregated drawing spaces for different audience members.

Ultimately, we adopted a scrolling shape-based drawing system, which provided enough structure to avoid mere novelty, struck a balance between prescription and free improvisation for the musicians, and helped define distinct identities for each participant based on their designed shape. Our drawing widget (Figure 8) is a horizontally scrolling score, shared among all audience participants. The canvas shows all shapes drawn by all users, maintaining time and horizontal position

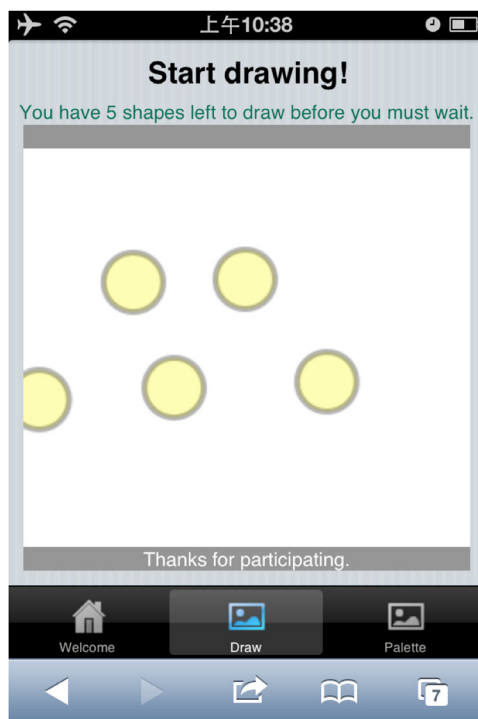


Figure 8. The scrolling score drawing and display widget for *Sketching*.

synchronization among connected devices. The shape palette (Figure 9) lets participants design the object to be drawn in the drawing space, customizing its shape, color, border width, size and opacity. Each user begins with a randomly configured shape.

Musicians read the score from a phone or tablet on their music stand or from a video projection. (The drummer does not play from the score but instead keeps steady time.) Time is mapped to the horizontal axis, with shapes being played as they touch the left border of the screen, and the vertical axis maps to pitch, with the exact pitch mappings left to the musicians' discretion based on the chord progression of the tune and their own musical sensibilities. Each type of shape (e.g. circle, triangle and square) is played by a different musician in the ensemble. The other qual-

ities of the shape suggest musical parameters such as articulation, volume, sustain and mood (Figure 9). Larger shapes are intended to be musical motives, not single notes.

Our Max/MSP software merely configures the mobile phone interface during performance, controlling options such as which shapes are available for drawing (and thus which instruments are playing), text messages displayed to audience participants, the scroll rate of the score and a count-down timer.

4.2.2 Guiding participation at scale

We presented *Sketching* in two performances with the Georgia Tech jazz combo in spring 2013 at two different venues: the larger, more formal First Center for the Arts at Georgia Tech and the smaller, more club-like Under the Couch performance space. Table 2 shows overall participation statistics across the two performances.

In the first performance at the First Center, the system was technically robust and audience participants found the interface intuitive to use, but the high density of participation was problematic, suggesting the need for additional mechanisms to guide and limit participation. Far more people participated than we had expected (about 40% of audience members), and they participated quite actively. Since there was no cap on participation, the score quickly became saturated with shapes (Figure 10), resembling a stained-glass window that was virtually impossible for the musicians to read. Therefore, the audience input essentially failed to influence the musicians due to its high density. One participant noted that “it was a mass of shapes and I felt I had to add more to keep up, which probably just made it even more jumbled”, while others complained that “the screen just looked like noise” and that “everybody just spammed their shape over and over again”. A small number of users were responsible for a majority of the shapes (the average shapes per user is triple the median).

For the second performance, we added new mechanisms to dynamically manage density by imposing drawing caps on participants. Participants were limited to drawing a fixed number of

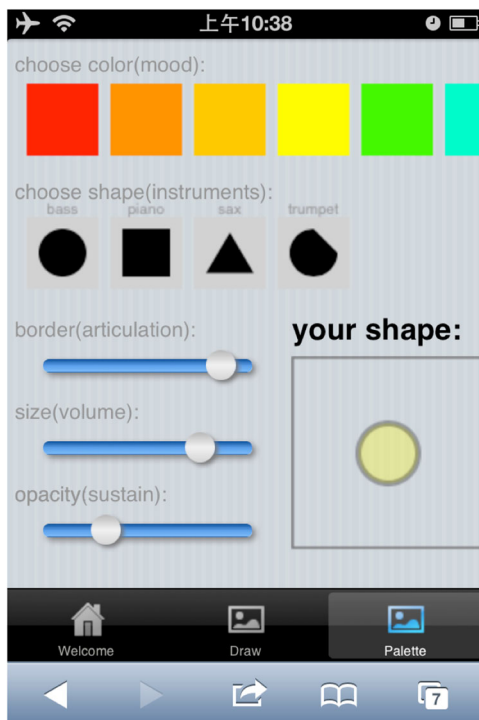


Figure 9. The shape palette widget for *Sketching*. The labels for each element of the interface can be configured in the Max/MSP patch. For the two performances we discuss, we mapped color, shape, border, size and opacity to the desired mood, instrument, articulation, volume and sustain, respectively.

	24 February First Center	17 April Under the Couch
No. of audience members	163	31
No. of shapes drawn	19,722	1,479
Avg. no. of shapes per user	120	47
Med. no. of shapes per user	42	35
Estimated attendance	~400	~90
Participation rate (%)	41%	34%

Table 2. Audience participation statistics from two live performances of Sketching.

Note: The participation rates are approximate values since no official attendance figures are available for either performance.

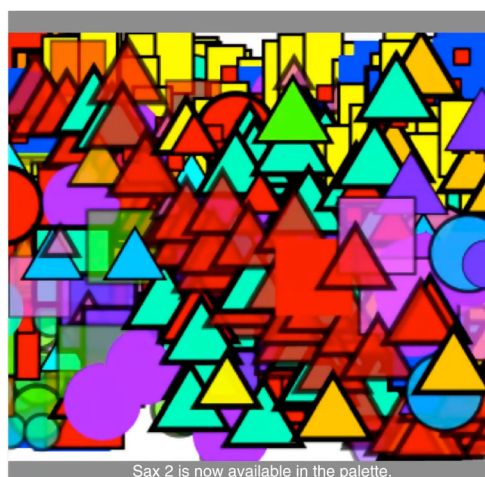


Figure 10. An example of the score created during the first performance of Sketching, with overly high density.

shapes (e.g. 10) per time period (e.g. a minute), and the widget showed participants their shape allocation or, if they ran out, how much longer they had to wait (Figure 11). This paradigm was directly inspired by other collaborative drawing tools (Edmunds 2006). The Max/MSP patch enabled dynamic modification of the shape limit and time period, which we used both to respond to the dynamics of a particular performance and to deliberately effect density changes to create a more directed, less static musical sensibility. This resulted in more readable scores in the second performance and eliminated most of the rapid-drawing outliers, as the average number of shapes per person approached the median.

The musical outcome of the first performance also exhibited problems with instrumentation and



Figure 11. An example of the score created during the second performance, with limits on overall density by enforcing turn-taking.

texture. Since participants were drawing different shapes, which represented different instruments, onto the score at the same time, all instruments were improvising together throughout. This was a poor match for how most jazz combos typically improvise, with musicians taking turns. We decided to dynamically enable and disable shapes from the drawing palette during the performance and to display text messages on the score when new shapes were made available. This enabled us to add instruments into the texture one by one during the second performance,

not only creating a more compelling textural structure but also creating a more legible mapping from score to resulting music, since each new shape/instrument was added individually.

4.2.3 Discussion

Sketching, like TeamWork, demonstrated the technical robustness of the massMobile framework at scale and afforded the opportunity to iteratively improve interaction designs in terms of interface, mappings and techniques to guide participation. Despite differences in context and structural complexity between the two projects, there were pronounced similarities. Both projects converged on X/Y designs for audience input and data display. Both projects also actively guided participants in their participation, whether through a mascot holding signs, text messages within the interface or changes to available options. And with both projects, participation numbers and density were a key design driver. In the case of TeamWork, this led to an overly cautious initial design for data display that we later discarded, while in the case of Sketching, this led to density problems in the initial design that we addressed through limits on participant activity rates.

5 Conclusion

In this paper, we have considered large-scale participatory musical performances in the context of our massMobile framework and two projects realized with it, TeamWork for an athletic band at sporting events and Sketching for a jazz combo in traditional concert venues. Within this context, we have focused on exploring (a) what technical designs provide robust support; (b) what interfaces and mappings can be effective and (c) what strategies can effectively focus and guide participation.

With respect to technical design, we found that a standards-based framework that employs a cloud-based architecture coupled with a browser-based mobile client can support participatory applications robustly at scale with zero reliance on venue-specific support, though at the expense of latency and jitter. We realize this runs counter

to the approach of many other systems (e.g. Schnell et al. 2015; Wyse 2015) that use dedicated local Wi-Fi routers and servers, but given the strong current state of cellular and wireless infrastructure, we have consistently seen wide-area architectures such as ours thrive in venues where local-area solutions fail. This does come with the caveat that in extremely large venues (such as football stadiums) or in remote or underground venues, cellular networks may be unavailable or overloaded.

With respect to effective interfaces and mappings, our work has converged on an X/Y style interface for both input and visualization. This follows both from X/Y pads that are ubiquitous in commercial music controllers and from a variety of participatory music projects that have employed this interface (Metois and Back 1996; Barbosa 2003). The critical challenge in employing this interface at scale is in event density and data display, considered through the lens of information theory (Galanter 2003). When all data points are displayed, participants can see a complete picture of activity, including their own, in the result, but if the display becomes too dense, then little useful information flows. By limiting the density of individual participants' activities, not only can a full-data approach be used at greater scale, but participants can also be nudged to reflect more on each action they take.

With respect to effective techniques to guide participation, these projects forced us to reconsider our esthetic biases toward self-organizing participatory systems and to implement techniques to overtly suggest participatory actions to audiences. This was partly due to scale—especially in the case of TeamWork, where participants were spread across an 8,600-seat basketball arena—and partly due to the anonymous nature of participation by mobile phone in contrast to other participation modes such as dancing, singing or waving light sticks. By employing charismatic leaders such as a team's mascot, sending suggestions via text display or enabling/disabling interface features during the performance, we made it more readily possible for these audiences to work collectively to effect substantial musical influence.

Acknowledgements

massMobile is available at <http://distributedmusic.gatech.edu/jason/research/massmobile.html>. A video recording of one of the TeamWork performances can be viewed at <http://distributedmusic.gatech.edu/jason/music/teamwork-2013-for-athletic/materials/performance-video.html>. A video recording of the second Sketching performance can be viewed at <http://distributedmusic.gatech.edu/jason/music/sketching-2013-for-improvis/materials/performance-video.html>. Thanks to Chip Crotts and Chris Moore for their generosity and open-mindedness in leading their ensembles in collaboration with us.

Funding

This work was supported in part by the National Science Foundation under grant NSF CreativeIT 0855758.

References

- Allison, Jesse, Yemin Oh, and Benjamin Taylor. 2013. "Nexus: Collaborative Performance for the Masses, Handling Instrument Interface Distribution Through the Web." *Proceedings of the New Interfaces for Musical Expression Conference*. http://aimlab.kaist.ac.kr/nime2013/program/papers/day1/paper1/287/287_Paper.pdf.
- Barbosa, Álvaro. 2003. "Displaced Soundscapes: A Survey of Network Systems for Music and Sonic Art Creation." *Leonardo Music Journal* 13: 53–59.
- Bregler, Christoph, Clothilde Castiglia, Jessica DeVincenzo, Roger Luke DuBois, Kevin Feeley, Tom Igoe, Jonathan Meyer, et al. 2005. "Squidball: An Experiment in Large-Scale Motion Capture and Game Design." In *Proceedings of the First International Conference on Intelligent Technologies for Interactive Entertainment*, 23–33. INTETAIN'05. Berlin: Springer-Verlag. doi:10.1007/11590323_3.
- Brown, Earle. 1999. "Transformations and Developments of a Radical Aesthetic." *Current Musicology* 67 (68): 39–57.
- Bryan-Kinns, N., and P. G. T. Healey. 2004. "Daisyphone: Support for Remote Music Collaboration." In *Proceedings of the 2004 Conference on New Interfaces for Musical Expression*, 27–30. NIME '04. Singapore: National University of Singapore. <http://dl.acm.org/citation.cfm?id=1085884.1085890>.
- Camurri, Antonio, Paolo Coletta, Giovanna Varni, and Simone Ghisio. 2007. "Developing Multimodal Interactive Systems with EyesWeb XML." In *Proceedings of the 7th International Conference on New Interfaces for Musical Expression*, 305–308. ACM. <http://dl.acm.org/citation.cfm?id=1279806>.
- Carpenter, Loren, and Rachel Carpenter. 1999. "Audience Participation." In *Ars Electronica: Facing the Future*, edited by Timothy Druckrey, 395–396. Cambridge, MA: MIT Press.
- Dahl, Luke, Jorge Herrera, and Carr Wilkerson. 2011. "Tweetdreams: Making Music with the Audience and the World Using Real-Time Twitter Data." *Proceedings of the International Conference on New Interfaces for Musical Expression (NIME)*, Oslo, Norway, May 30–June 1, 272–275.
- Dannenberg, Roger B., and Joseph Bates. 1995. "A Model for Interactive Art." <http://repository.cmu.edu/compsci/481/>.
- Drummond, Jon. 2009. "Understanding Interactive Systems." *Organised Sound* 14 (2): 124–33. doi:10.1017/S1355771809000235.
- Edmunds, Peter. 2006. "SwarmSketch." PDC, Trento, Italy, August 1–5, 21–22.
- Feldmeier, Mark, and Joseph A. Paradiso. 2007. "An Interactive Music Environment for Large Groups with Giveaway Wireless Motion Sensors." *Computer Music Journal* 31 (1): 50–67.
- Freeman, Jason, and Mark Godfrey. 2010. "Creative Collaboration Between Audiences and Musicians in Flock." *Digital Creativity* 21: 85–99. doi:10.1080/14626261003786244.
- Galanter, Philip. 2003. "What Is Generative Art? Complexity Theory as a Context for Art Theory." GA2003—6th Generative Art Conference, Milan, Italy, December 10–13.
- Harries, Guy. 2013. "'The Open Work': Ecologies of Participation." *Organised Sound* 18 (1): 3–13.
- Hasse, Jean. 1986. *Moths*. Euclid, OH: Visible Music.
- Hindle, Abram. 2013. "Swarmed: Captive Portals, Mobile Devices, and Audience Participation in Multi-user Music Performance." *Proceedings of the 13th International Conference on New Interfaces for Musical Expression*, 174–179. http://aimlab.kaist.ac.kr/nime2013/program/papers/day2/paper4/62/62_Paper.pdf.
- Jordà, Sergi. 2003. "Interactive Music Systems for Everyone: Exploring Visual Feedback as a Way for Creating More Intuitive, Efficient and Learnable Instruments."

- Proceedings of the Stockholm Music Acoustics Conference (SMAC03)*, Stockholm, Sweden. <http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.14.3389&rep=rep1&type=pdf>.
- Jorda, Sergi. 2007. "Interactivity and Live Computer Music." In *The Cambridge Companion to Electronic Music*, edited by Nick Collins and Julio d'Escrivan, 89–106. Cambridge: Cambridge University Press.
- Kim, Haeyoung. 2011. "Moori: Interactive Audience Participatory Audio-Visual Performance." In *Proceedings of the 8th ACM Conference on Creativity and Cognition*, 437–438. ACM. <http://dl.acm.org/citation.cfm?id=2069737>.
- Lee, Sang Won, and Jason Freeman. 2013. "Echobo: A Mobile Music Instrument Designed for Audience to Play." *Proceedings of the New Interfaces for Musical Expression Conference (NIME 2013)*, Seoul, Korea, May 27–30.
- Lohner, Henning. 1986. "The UPIC System: A User's Report." *Computer Music Journal* 10 (4): 42–49.
- Maynes-Aminzade, Dan, Randy Pausch, and Steve Seitz. 2002. "Techniques for Interactive Audience Participation." In *Proceedings of the 4th IEEE International Conference on Multimodal Interfaces*, 15. ICMI '02. Washington, DC: IEEE Computer Society. doi:10.1109/ICMI.2002.1166962.
- McAllister, Graham, Michael Alcorn, and Philip Strain. 2004. "Interactive Performance with Wireless PDAs." *Proceedings of the 2004 International Computer Music Conference*, Miami, Florida, November 1–6, 702–705.
- Metois, Eric, and Maribeth Back. 1996. "BROWeb: An Interactive Collaborative Auditory Environment on the World Wide Web." *Proceedings of the Third International Conference on Auditory Display*, Palo Alto, California, November 4–6, 105–110.
- Oh, Jieun, and Ge Wang. 2011. "Audience-Participation Techniques Based on Social Mobile Computing." In *Proceedings of the 2011 International Computer Music Conference*. Huddersfield: MPublishing, University of Michigan Library. <https://crma.stanford.edu/groups/mcd/publish/files/2011-icmc-audience.pdf>.
- Paul, Christiane, and Christian Werner. 2003. *Digital Art*. London: Thames & Hudson. http://www.flong.com/storage/pdf/press/2003_paul_digitalart.pdf.
- Pitaru, Amit. 2005. "Amit Pitaru." In *ACM SIGGRAPH 2005 Electronic Art and Animation Catalog*, 130–131. ACM. <http://dl.acm.org/citation.cfm?id=1086121>.
- Roberts, Charles, and Tobias Hollerer. 2011. "Composition for Conductor and Audience: New Uses for Mobile Devices in the Concert Hall." In *Proceedings of the 24th Annual ACM Symposium Adjunct on User Interface Software and Technology*, 65–66. ACM. <http://dl.acm.org/citation.cfm?id=2046425>.
- Rokeby, David. 1998. "The Construction of Experience: Interface as Content." In *Digital Illusion: Entertaining the Future with High Technology*, edited by Clark Dodsworth, Jr., 27–48. New York, NY: ACM Press.
- Rowe, Robert. 1992. *Interactive Music Systems: Machine Listening and Composing*. Cambridge, MA: MIT Press.
- Schacher, Jan. 2008. "Action and Perception in Interactive Sound Installations: An Ecological Approach." *Proceedings of the New Interfaces for Musical Expression, NIME 2009 Conference*. Pittsburgh, PA. http://www.zhdk.ch/fileadmin/data_subsites/data_icst/Forschung/MGM/Publikationen/MGM_Nime09_Action_Perception.pdf.
- Schnell, Norbert, Sébastien Robaszkiewicz, Frederic Bevilacqua, and Diemo Schwarz. 2015. "Collective Sound Checks: Exploring Intertwined Sonic and Social Affordances of Mobile Web Applications." In *Proceedings of the Ninth International Conference on Tangible, Embedded, and Embodied Interaction*, 685–690. ACM. <http://dl.acm.org/citation.cfm?id=2688808>.
- Stockhausen, Karlheinz. 1957. *Klavierstück XI*. No. 12654 LW. Vienna: Universal Edition.
- Thorington, Helen. 2005. "Breaking Out: The Trip Back." *Contemporary Music Review* 24 (6): 445–458.
- Traub, Peter. 2005. "Sounding the Net: Recent Sonic Works for the Internet and Computer Networks." *Contemporary Music Review* 24 (6): 459–481. doi:10.1080/07494460500296136.
- Wechsler, Robert, Frieder Weiss, and Peter Dowling. 2004. "EyeCon: A Motion Sensing Tool for Creating Interactive Dance, Music, and Video Projections." *Proceedings of the AISB 2004 COST287-ConGAS Symposium on Gesture Interfaces for Multimedia Systems*, Leeds, UK, March 29–April 1, 74–79.
- Weinberg, Gil. 2005. "Interconnected Musical Networks: Toward a Theoretical Framework." *Computer Music Journal* 29 (2): 23–39.
- Weitzner, Nathan, Jason Freeman, Yan-Ling Chen, and Stephen Garrett. 2013. "massMobile: Towards a Flexible Framework for Large-Scale Participatory Collaborations in Live Performances." *Organised*

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Sound 18 (Special Issue 1): 30–42. doi:10.1017/S1355771812000222.

Weitzner, Nathan, Jason Freeman, Stephen Garrett, and Y. Chen. 2012. “massMobile—an Audience Participation Framework.” In *Proceedings of the New Interfaces for Musical Expression Conference*, Ann Arbor, MI, May 21–23.

Wyse, Lonce. 2015. “Spatially Distributed Sound Computing and Rendering Using the Web Audio Platform.” *Proceedings of the 1st Annual Web Audio Conference*, Paris, January 26–28.

Zorn, John. 1995. *John Zorn’s Cobra Live at the Knitting Factory*. New York, NY: Knitting Factory Works.

Jason Freeman is an Associate Professor of Music at Georgia Tech, where he teaches music technology, composes music that breaks down the barriers between composers, performers and listeners, and develops novel educational technologies that combine music and computer science.

Shaoduo Xie received an MS in music technology from Georgia Tech in 2014 and is the co-founder of Ample Sound, an audio plug-in developer.

Takahiko Tsuchiya is a PhD candidate in music technology at Georgia Tech, where his research interests include data sonification, live coding, composition and STEAM education.

Weibin Shen received an MS in music technology from Georgia Tech in 2013 and is currently a software engineer at Apple.

Yan-Ling Chen received an MS in music technology from Georgia Tech in 2013 and is currently a mobile application developer for Sweetspot.

Nathan Weitzner received an MS in music technology from Georgia Tech in 2012 and is currently a PhD candidate in media arts and technology at the University of California, Santa Barbara.